

Tutorial 10: Instrumental Variables and 2SLS (MROZ) — v2

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Contents

Overview	2
Module 1: What is <code>AER::ivreg()</code>? (first-time tool)	3
Why do we need IV / 2SLS?	3
<code>ivreg()</code> formula syntax	3
1. Data: MROZ working women (non-missing wages)	3
What the problem is asking	3
Step-by-step plan	4
Common pitfalls	4
Load and filter	4
2. Baseline OLS: return to education	5
3. IV (just-identified): instrument educ with <code>fatheduc</code>	5
4. Manual IV estimator (covariance ratio)	6
5. Manual 2SLS (intuition) and why naive SE are wrong	6
5.1 First stage	6
5.2 Second stage (naive)	6
6. Richer 2SLS model (controls + multiple instruments)	7
Module 2: The <code>I()</code> function in formulas (first-time tool in this doc)	7
7. Compare OLS vs IV education slopes	8
8. Getting the “correct” standard errors in IV	8
8.1 What the problem is asking	8
8.2 Step-by-step plan	8
8.3 Common pitfalls	8
How to read the two outputs (conventional IV SE vs robust IV SE)	9

Module 3: What are <code>vcovHC()</code> and <code>coeftest()</code> ? (first-time tools)	9
<code>sandwich::vcovHC()</code>	9
<code>lmtest::coeftest()</code>	9
8.4 Robust IV standard errors (HC1)	9
9. Hausman-type test for endogeneity (DWH / Wu–Hausman)	10
9.1 What the problem is asking	10
9.2 Step-by-step plan	10
9.3 Common pitfalls	10
Module 4: What does <code>summary(ivreg_model, diagnostics = TRUE)</code> do?	10
10. Checking IV strength for <code>iv2</code> (MROZ): first-stage F statistics	11
10.1 What the problem is asking	11
10.2 Step-by-step plan	11
10.3 Common pitfalls	12
Module 5: What is <code>car::linearHypothesis()</code> ? (first-time tool in this doc)	12
10.4 First stage + excluded instruments joint F	12
Wrap-up	13

Overview

This tutorial builds an end-to-end IV workflow (2SLS) using the **MROZ** dataset:

- 1) How to estimate **2SLS in R** (`AER::ivreg`)
- 2) How to report **correct standard errors** (including **robust** IV standard errors)
- 3) How to run a **Hausman-type endogeneity test** (Durbin–Wu–Hausman / Wu–Hausman)
- 4) How to assess **instrument strength** (first-stage F statistics; “weak IV” diagnostics)

We use pipes (`%>%`) for clarity.

```
if (!requireNamespace("wooldridge", quietly = TRUE)) install.packages("wooldridge")
if (!requireNamespace("dplyr", quietly = TRUE)) install.packages("dplyr")
if (!requireNamespace("AER", quietly = TRUE)) install.packages("AER")

# For robust standard errors and formatted tests:
if (!requireNamespace("sandwich", quietly = TRUE)) install.packages("sandwich")
if (!requireNamespace("lmtest", quietly = TRUE)) install.packages("lmtest")

# For F-tests on excluded instruments:
if (!requireNamespace("car", quietly = TRUE)) install.packages("car")

library(wooldridge)
library(dplyr)
library(AER)
```

```
library(sandwich)
library(lmtest)
library(car)
```

Module 1: What is `AER::ivreg()`? (first-time tool)

`ivreg()` fits an **instrumental variables** regression (2SLS under the hood).

Why do we need IV / 2SLS?

OLS requires the regressor to be exogenous:

$$E(u \mid x) = 0.$$

If `educ` is correlated with the error term in the wage equation (e.g., ability/omitted factors), OLS is biased. An **instrument** z must satisfy:

1. **Relevance:** $\text{Cov}(z, x) \neq 0$
2. **Exogeneity:** $\text{Cov}(z, u) = 0$

`ivreg()` formula syntax

```
ivreg(y ~ x1 + x2 + ... | z1 + z2 + ...)
```

- Left of `|` = regressors in the structural equation (endogenous + exogenous)
- Right of `|` = instruments (must include all exogenous regressors as their own instruments, plus excluded instruments)

Example (one endogenous regressor `educ`, one instrument `fatheduc`):

```
ivreg(lwage ~ educ | fatheduc)
```

1. Data: MROZ working women (non-missing wages)

What the problem is asking

We want the **causal** effect of education on log wage. We will:

1. Estimate the OLS return to education.
2. Re-estimate using IV, treating `educ` as endogenous.
3. Compare OLS vs IV estimates.
4. Understand the IV estimator in the just-identified case (covariance ratio).
5. Understand 2SLS (first stage + second stage) and why naive second-stage SE are wrong.
6. Estimate a richer 2SLS model with controls and multiple instruments.
7. Add diagnostics: robust IV SE, Hausman-type endogeneity test, and first-stage strength checks.

Step-by-step plan

- 1) Load `mroz` and keep observations with non-missing `lwage`.
- 2) Run OLS: `lm(lwage ~ educ)`.
- 3) Run IV: `ivreg(lwage ~ educ | fatheduc)`.
- 4) Compute IV slope manually: $\hat{\beta}_1 = \text{Cov}(z, y) / \text{Cov}(z, x)$.
- 5) Do 2SLS by hand: first stage `educ ~ fatheduc`, second stage `lwage ~ educ` (for intuition).
- 6) Run the richer model:
$$lwage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u,$$
instrumenting `educ` with parents' education.
- 7) Add **robust IV SE**, **Hausman-type test**, and **first-stage F** strength checks.

Common pitfalls

- **Instrument validity:** relevance is not enough; exogeneity is the hard part (cannot be fully tested).
- **Naive 2SLS standard errors:** `lm(lwage ~ educ)` reports invalid SE.
- **Forgetting exogenous controls must appear on both sides of |** in `ivreg()`.
- **Log wages interpretation:** 0.10 in log points is roughly a 10% wage change (approx.).

Load and filter

```
data("mroz")

df <- mroz %>%
  as_tibble() %>%
  filter(!is.na(lwage))

df %>% summarise(n = n())
```

```
## # A tibble: 1 x 1
##       n
##   <int>
## 1    428
```

```
df %>% select(lwage, educ, exper, fatheduc, motheduc) %>% summary()
```

```
##      lwage      educ      exper      fatheduc
##  Min.   :-2.0542  Min.    : 5.00  Min.    : 0.00  Min.    : 0.000
## 1st Qu.: 0.8165  1st Qu.:12.00  1st Qu.: 7.00  1st Qu.: 7.000
## Median : 1.2476  Median :12.00  Median :12.00  Median : 7.000
## Mean   : 1.1902  Mean    :12.66  Mean    :13.04  Mean    : 8.988
## 3rd Qu.: 1.6036  3rd Qu.:14.00  3rd Qu.:18.00  3rd Qu.:12.000
## Max.    : 3.2189  Max.    :17.00  Max.    :38.00  Max.    :17.000
##      motheduc
##  Min.    : 0.000
## 1st Qu.: 7.000
```

```
## Median :10.000
## Mean   : 9.516
## 3rd Qu.:12.000
## Max.    :17.000
```

2. Baseline OLS: return to education

```
ols1 <- lm(lwage ~ educ, data = df)
summary(ols1)
```

```
##
## Call:
## lm(formula = lwage ~ educ, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.10256 -0.31473  0.06434  0.40081  2.10029
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.1852     0.1852  -1.000   0.318
## educ          0.1086     0.0144   7.545 2.76e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.68 on 426 degrees of freedom
## Multiple R-squared:  0.1179, Adjusted R-squared:  0.1158
## F-statistic: 56.93 on 1 and 426 DF,  p-value: 2.761e-13
```

3. IV (just-identified): instrument educ with fatheduc

```
iv1 <- ivreg(lwage ~ educ | fatheduc, data = df)
summary(iv1)
```

```
##
## Call:
## ivreg(formula = lwage ~ educ | fatheduc, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0870 -0.3393  0.0525  0.4042  2.0677
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.44110     0.44610   0.989   0.3233
## educ         0.05917     0.03514   1.684   0.0929 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6894 on 426 degrees of freedom
## Multiple R-Squared:  0.09344, Adjusted R-squared:  0.09131
## Wald test: 2.835 on 1 and 426 DF,  p-value: 0.09294
```

4. Manual IV estimator (covariance ratio)

In a just-identified model with one instrument z and one endogenous regressor x , the IV slope is:

$$\hat{\beta}_{1,IV} = \frac{\widehat{\text{Cov}}(z, y)}{\widehat{\text{Cov}}(z, x)}.$$

```
beta1hat_iv <- with(df, cov(fatheduc, lwage) / cov(fatheduc, educ))
beta0hat_iv <- with(df, mean(lwage) - beta1hat_iv * mean(educ))

c(beta0hat_iv = beta0hat_iv, beta1hat_iv = beta1hat_iv)
```

```
## beta0hat_iv beta1hat_iv
## 0.44110341 0.05917348
```

```
coef(iv1)
```

```
## (Intercept)      educ
## 0.44110341 0.05917348
```

5. Manual 2SLS (intuition) and why naive SE are wrong

5.1 First stage

```
fs <- lm(educ ~ fatheduc, data = df)
summary(fs)
```

```
##
## Call:
## lm(formula = educ ~ fatheduc, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -8.4704 -1.1231 -0.1231  0.9546  5.9546
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  10.23705    0.27594   37.099  <2e-16 ***
## fatheduc      0.26944    0.02859    9.426  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.081 on 426 degrees of freedom
## Multiple R-squared:  0.1726, Adjusted R-squared:  0.1706
## F-statistic: 88.84 on 1 and 426 DF, p-value: < 2.2e-16
```

```
df <- df %>% mutate(educhat = fitted(fs))
```

5.2 Second stage (naive)

```
ss_naive <- lm(lwage ~ educhat, data = df)
summary(ss_naive)
```

```
##
## Call:
## lm(formula = lwage ~ educhat, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2126 -0.3763  0.0563  0.4173  2.0604
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.44110     0.46711   0.944   0.346
## educhat      0.05917     0.03680   1.608   0.109
##
## Residual standard error: 0.7219 on 426 degrees of freedom
## Multiple R-squared:  0.006034,    Adjusted R-squared:  0.003701
## F-statistic: 2.586 on 1 and 426 DF,  p-value: 0.1086
```

Key point: the coefficient is typically close to IV, but the **standard errors are not valid** for 2SLS because `educhat` is estimated. Use `ivreg()` for correct 2SLS inference.

6. Richer 2SLS model (controls + multiple instruments)

We estimate:

$$lwage = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u,$$

treating `educ` as endogenous, and using `fatheduc` and `motheduc` as excluded instruments.

Module 2: The `I()` function in formulas (first-time tool in this doc)

In R formulas, `I()` means “interpret this literally as arithmetic.”

For example, `I(exper^2)` includes the variable $exper^2$ in the regression.

```
iv2 <- ivreg(
  lwage ~ educ + exper + I(exper^2) |
  exper + I(exper^2) + fatheduc + motheduc,
  data = df
)

summary(iv2)
```

```
##
## Call:
## ivreg(formula = lwage ~ educ + exper + I(exper^2) | exper + I(exper^2) +
##      fatheduc + motheduc, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
```

```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0481003  0.4003281   0.120  0.90442
## educ        0.0613966  0.0314367   1.953  0.05147 .
## exper       0.0441704  0.0134325   3.288  0.00109 **
## I(exper^2)  -0.0008990  0.0004017  -2.238  0.02574 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6747 on 424 degrees of freedom
## Multiple R-Squared:  0.1357, Adjusted R-squared:  0.1296
## Wald test: 8.141 on 3 and 424 DF, p-value: 2.787e-05
```

7. Compare OLS vs IV education slopes

```
c(
  ols_educ = coef(ols1)["educ"],
  iv1_educ = coef(iv1)["educ"],
  iv2_educ = coef(iv2)["educ"]
)
```

```
## ols_educ.educ iv1_educ.educ iv2_educ.educ
##      0.10864866      0.05917348      0.06139663
```

8. Getting the “correct” standard errors in IV

8.1 What the problem is asking

You need standard errors that are valid for IV/2SLS. Two practical layers:

- 1) **Correct 2SLS SE**: do not use the naive second-stage SE; use `ivreg()` output.
- 2) **Robust 2SLS SE** (recommended in applied work): allow heteroskedasticity using `sandwich`.

8.2 Step-by-step plan

- 1) Use `summary(iv2)` for conventional (homoskedastic) 2SLS SE.
- 2) Compute robust VCOV with `sandwich::vcovHC(iv2, type="HC1")`.
- 3) Print coefficient tests with robust SE using `lmtest::coeftest()`.

8.3 Common pitfalls

- Robust SE change **t statistics/p-values**, not coefficients.
- HC0 vs HC1: HC1 is a common small-sample correction.
- Don’t mix OLS robust SE with IV models: use `vcovHC()` on the **ivreg** object.

How to read the two outputs (conventional IV SE vs robust IV SE)

You will often see two slightly different standard errors for the *same* IV coefficient estimates:

- `summary(iv2)` reports **conventional 2SLS standard errors**, which rely on the homoskedasticity assumption.
- `coeftest(iv2, vcov. = vcovHC(iv2, ...))` reports **robust (heteroskedasticity-consistent) IV standard errors**.

What is being compared?

Not “IV vs non-IV.” The coefficient estimates $\hat{\beta}$ are the same IV/2SLS estimates in both cases.

What changes is the **variance estimator** used for inference (SE, t-stat, p-value).

Which is more accurate? - If the error variance is truly constant (homoskedastic), conventional SE can be more efficient (often smaller). - If heteroskedasticity is present (common in practice), robust SE are typically more reliable for inference.

A common applied default is to **report robust IV SE (HC1)** unless the assignment explicitly asks for the homoskedastic version.

Module 3: What are `vcovHC()` and `coeftest()`? (first-time tools)

`sandwich::vcovHC()`

Computes a heteroskedasticity-consistent covariance matrix for coefficient estimates.
You can apply it to `lm` and also to `ivreg` objects.

`lmtest::coeftest()`

Prints coefficient tests using a user-supplied covariance matrix (e.g., robust VCOV).

8.4 Robust IV standard errors (HC1)

```
robust_vcov_iv2 <- vcovHC(iv2, type = "HC1")

# Robust coefficient table (large-sample reference by default)
coeftest(iv2, vcov. = robust_vcov_iv2)

##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.04810031  0.42979771  0.1119 0.910945
## educ        0.06139663  0.03333859  1.8416 0.066231 .
## exper       0.04417039  0.01554638  2.8412 0.004711 **
## I(exper^2)  -0.00089897  0.00043008 -2.0902 0.037193 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Optional: compare conventional vs robust SE:

```
se_conventional <- sqrt(diag(vcov(iv2)))
se_robust <- sqrt(diag(robust_vcov_iv2))

cbind(se_conventional = se_conventional, se_robust = se_robust)
```

```
##          se_conventional    se_robust
## (Intercept)    0.4003280776 0.4297977133
## educ          0.0314366956 0.0333385881
## exper         0.0134324755 0.0155463781
## I(exper^2)    0.0004016856 0.0004300837
```

9. Hausman-type test for endogeneity (DWH / Wu–Hausman)

9.1 What the problem is asking

We want to test whether `educ` behaves as **endogenous** in the wage equation.

A Hausman-type test compares OLS and IV in a structured way. In `ivreg`, this is available as a **Wu–Hausman** (Durbin–Wu–Hausman) diagnostic.

9.2 Step-by-step plan

- 1) Estimate `iv2`.
- 2) Call `summary(iv2, diagnostics = TRUE)`.
- 3) Read the **Wu–Hausman** test:
 - If p-value is small: evidence that OLS is inconsistent (endogeneity), IV preferred.
 - If p-value is large: no strong evidence against exogeneity (OLS may be OK).

9.3 Common pitfalls

- This tests **endogeneity of the instrumented regressor(s)**, not instrument validity.
- A weak instrument can make IV noisy; endogeneity tests can have low power.
- “Fail to reject” does not prove exogeneity; it means insufficient evidence.

Module 4: What does `summary(ivreg_model, diagnostics = TRUE)` do?

For `ivreg` objects, setting `diagnostics = TRUE` prints standard IV diagnostics, typically including:

- **Weak instruments** (first-stage relevance, reported via an F-type statistic)
- **Wu–Hausman** (endogeneity / Hausman-type test)
- **Sargan** (overidentification test when overidentified; relies on homoskedasticity)

```
summary(iv2, diagnostics = TRUE)
```

```
##
## Call:
## ivreg(formula = lwage ~ educ + exper + I(exper^2) | exper + I(exper^2) +
##       fatheduc + motheduc, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0986 -0.3196  0.0551  0.3689  2.3493
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.0481003  0.4003281   0.120  0.90442
## educ        0.0613966  0.0314367   1.953  0.05147 .
## exper       0.0441704  0.0134325   3.288  0.00109 **
## I(exper^2)  -0.0008990  0.0004017  -2.238  0.02574 *
##
## Diagnostic tests:
##               df1 df2 statistic p-value
## Weak instruments    2 423    55.400 <2e-16 ***
## Wu-Hausman         1 423     2.793  0.0954 .
## Sargan             1 NA      0.378  0.5386
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6747 on 424 degrees of freedom
## Multiple R-Squared:  0.1357, Adjusted R-squared:  0.1296
## Wald test: 8.141 on 3 and 424 DF, p-value: 2.787e-05
```

10. Checking IV strength for iv2 (MROZ): first-stage F statistics

In the richer MROZ IV model (iv2), we treat **educ** as endogenous and use **parents' education** as excluded instruments:

- Endogenous regressor: **educ**
- Excluded instruments: **fatheduc**, **motheduc**
- Exogenous controls (also included as instruments): **exper**, **exper²**

So the relevant first stage for **iv2** is:

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 fatheduc + \pi_4 motheduc + v.$$

What we care about is whether the **excluded instruments** explain variation in **educ** *after controlling for* **exper** and **exper²**.

10.1 What the problem is asking

We want evidence that the excluded instruments (**fatheduc**, **motheduc**) are correlated with **educ** **conditional on** the exogenous controls.

A standard check is the **joint F test** in the first stage:

$$H_0 : \pi_3 = 0, \pi_4 = 0.$$

This is the “instrument strength” check most often referenced by the rule of thumb “first-stage $F > 10$ ” (very rough guideline).

10.2 Step-by-step plan

- 1) Fit the first-stage regression used by **iv2**:

$$educ = \pi_0 + \pi_1 exper + \pi_2 exper^2 + \pi_3 fatheduc + \pi_4 motheduc + v.$$

- 2) Test the joint null that the **excluded instruments** have zero coefficients:

$$H_0 : \pi_3 = 0, \pi_4 = 0.$$

- 3) Report the resulting F statistic and p-value.
4) Interpret: larger F means stronger partial correlation between instruments and **educ**.

10.3 Common pitfalls

- Using the overall model F instead of the **excluded-instruments** joint F.
- Forgetting to include exogenous controls in the first stage when they appear in the second stage.
- Treating “F > 10” as a guarantee (it is not; it is a rule of thumb).

Module 5: What is `car::linearHypothesis()`? (first-time tool in this doc)

`linearHypothesis()` conducts Wald/F tests of linear restrictions like:

$$H_0 : \pi_3 = 0, \pi_4 = 0.$$

It is an easy way to obtain the **joint F test** for excluded instruments.

10.4 First stage + excluded instruments joint F

```
fs2 <- lm(educ ~ exper + I(exper^2) + fatheduc + motheduc, data = df)
summary(fs2)
```

```
##
## Call:
## lm(formula = educ ~ exper + I(exper^2) + fatheduc + motheduc,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.8057 -1.0520 -0.0371  1.0258  6.3787
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.102640    0.426561  21.340 < 2e-16 ***
## exper         0.045225    0.040251   1.124  0.262
## I(exper^2)   -0.001009    0.001203  -0.839  0.402
## fatheduc      0.189548    0.033756   5.615 3.56e-08 ***
## motheduc      0.157597    0.035894   4.391 1.43e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.039 on 423 degrees of freedom
## Multiple R-squared:  0.2115, Adjusted R-squared:  0.204
## F-statistic: 28.36 on 4 and 423 DF,  p-value: < 2.2e-16
```

```
# Joint F test: excluded instruments are irrelevant
linearHypothesis(fs2, c("fatheduc = 0", "motheduc = 0"))
```

```
## Linear hypothesis test
##
## Hypothesis:
## fatheduc = 0
## motheduc = 0
##
## Model 1: restricted model
## Model 2: educ ~ exper + I(exper^2) + fatheduc + motheduc
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      425 2219.2
## 2      423 1758.6  2    460.64 55.4 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Wrap-up

- **2SLS in R:** `AER::ivreg()`
- **Correct SE:** use `ivreg()` (not naive second stage); add robust SE via `vcovHC()` + `coeftest()`
- **Hausman-type test:** `summary(iv2, diagnostics = TRUE)` provides Wu–Hausman (endogeneity)
- **Instrument strength:** check excluded-instruments joint F in the first stage (and the weak-instruments diagnostic)